

BABU BANARASI DAS

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SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

Library Management System Using MongoDB

Submitted to:

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(55)

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1. CREATE DATABASE

MongoDB Query:

use library

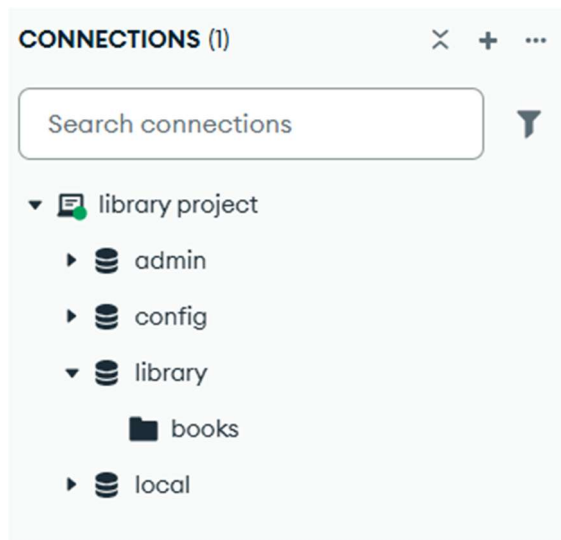
MongoDB uses the `use` command to create or switch to a database.

```
>_MONGOSH
> use("library")
< switched to db library
```

2. CREATE COLLECTION

MongoDB Query:

`db.createCollection("books")`



3. INSERT 20 STUDENTS

MongoDB Query:

```
db.books.insertMany([  
  
  {bookId:1,bookName:"Python Basics",author:"Mark  
Lutz",price:500,category:"Technology",quantity:10},  
  
  {bookId:2,bookName:"Java Programming",author:"Herbert  
Schildt",price:700,category:"Technology",quantity:8},  
  
  {bookId:3,bookName:"Data Science",author:"John  
Smith",price:800,category:"Technology",quantity:6},  
  
  {bookId:4,bookName:"Machine Learning",author:"Tom  
Mitchell",price:900,category:"Technology",quantity:5},  
  
  {bookId:5,bookName:"Atomic Habits",author:"James Clear",price:500,category:"Self  
Help",quantity:12},  
  
  {bookId:6,bookName:"The Alchemist",author:"Paulo  
Coelho",price:300,category:"Fiction",quantity:7},  
  
  {bookId:7,bookName:"Rich Dad Poor Dad",author:"Robert  
Kiyosaki",price:450,category:"Finance",quantity:9},  
  
  {bookId:8,bookName:"Ikigai",author:"Hector Garcia",price:350,category:"Self  
Help",quantity:11},  
  
  {bookId:9,bookName:"Harry Potter",author:"J.K.  
Rowling",price:600,category:"Fantasy",quantity:6},  
  
  {bookId:10,bookName:"Wings of Fire",author:"A.P.J. Abdul  
Kalam",price:250,category:"Biography",quantity:10},  
  
  {bookId:11,bookName:"Clean Code",author:"Robert  
Martin",price:850,category:"Technology",quantity:5},  
  
  {bookId:12,bookName:"C Programming",author:"Dennis  
Ritchie",price:400,category:"Technology",quantity:8},  
  
  {bookId:13,bookName:"Think and Grow Rich",author:"Napoleon  
Hill",price:300,category:"Self Help",quantity:9},
```

{bookId:14,bookName:"The Hobbit",author:"J.R.R. Tolkien",price:550,category:"Fantasy",quantity:7},

{bookId:15,bookName:"1984",author:"George Orwell",price:450,category:"Fiction",quantity:6},

{bookId:16,bookName:"SQL Fundamentals",author:"John Watson",price:650,category:"Technology",quantity:8},

{bookId:17,bookName:"Deep Work",author:"Cal Newport",price:500,category:"Self Help",quantity:10},

{bookId:18,bookName:"Atomic Structure",author:"R. Sharma",price:200,category:"Education",quantity:12},

{bookId:19,bookName:"Digital Marketing",author:"Philip Kotler",price:750,category:"Business",quantity:5},

{bookId:20,bookName:"Business Studies",author:"Poonam Gandhi",price:350,category:"Business",quantity:8},

{bookId:21,bookName:"Operating Systems",author:"Galvin",price:800,category:"Technology",quantity:6},

{bookId:22,bookName:"Computer Networks",author:"Andrew Tanenbaum",price:850,category:"Technology",quantity:5},

{bookId:23,bookName:"Database Systems",author:"Raghu Ramakrishnan",price:900,category:"Technology",quantity:4},

{bookId:24,bookName:"Artificial Intelligence",author:"Stuart Russell",price:950,category:"Technology",quantity:5},

{bookId:25,bookName:"The Psychology of Money",author:"Morgan Housel",price:600,category:"Finance",quantity:9},

{bookId:26,bookName:"Rich Dad Poor Dad 2",author:"Robert Kiyosaki",price:450,category:"Finance",quantity:7},

{bookId:27,bookName:"Sherlock Holmes",author:"Arthur Conan Doyle",price:300,category:"Fiction",quantity:6},

{bookId:28,bookName:"The Kite Runner",author:"Khaled Hosseini",price:400,category:"Fiction",quantity:5},

{bookId:29,bookName:"Harry Potter 2",author:"J.K. Rowling",price:650,category:"Fantasy",quantity:8},

{bookId:30,bookName:"The Power of Now",author:"Eckhart Tolle",price:550,category:"Self Help",quantity:10},

{bookId:31,bookName:"JavaScript Guide",author:"David Flanagan",price:700,category:"Technology",quantity:5},

{bookId:32,bookName:"Web Development",author:"Jon Duckett",price:800,category:"Technology",quantity:6},

{bookId:33,bookName:"HTML and CSS",author:"Jon Duckett",price:600,category:"Technology",quantity:8},

{bookId:34,bookName:"Python Crash Course",author:"Eric Matthes",price:750,category:"Technology",quantity:7},

{bookId:35,bookName:"Data Analytics",author:"James Evans",price:650,category:"Technology",quantity:6},

{bookId:36,bookName:"Statistics Basics",author:"S.C. Gupta",price:250,category:"Education",quantity:10},

{bookId:37,bookName:"Physics Fundamentals",author:"H.C. Verma",price:500,category:"Education",quantity:8},

{bookId:38,bookName:"Chemistry Guide",author:"O.P. Tandon",price:450,category:"Education",quantity:7},

{bookId:39,bookName:"Mathematics",author:"R.D. Sharma",price:550,category:"Education",quantity:9},

{bookId:40,bookName:"The Secret",author:"Rhonda Byrne",price:400,category:"Self Help",quantity:10},

{bookId:41,bookName:"Think Like a Monk",author:"Jay Shetty",price:600,category:"Self Help",quantity:8},

```
{bookId:42,bookName:"The Fault in Our Stars",author:"John  
Green",price:350,category:"Fiction",quantity:6},  
  
{bookId:43,bookName:"Pride and Prejudice",author:"Jane  
Austen",price:450,category:"Fiction",quantity:5},  
  
{bookId:44,bookName:"The Lord of the Rings",author:"J.R.R.  
Tolkien",price:900,category:"Fantasy",quantity:4},  
  
{bookId:45,bookName:"The Chronicles of Narnia",author:"C.S.  
Lewis",price:700,category:"Fantasy",quantity:6},  
  
{bookId:46,bookName:"Startup Ideas",author:"Peter  
Thiel",price:800,category:"Business",quantity:5},  
  
{bookId:47,bookName:"Marketing Management",author:"Philip  
Kotler",price:850,category:"Business",quantity:6},  
  
{bookId:48,bookName:"Economics Today",author:"Paul  
Samuelson",price:500,category:"Business",quantity:7},  
  
{bookId:49,bookName:"The Lean Startup",author:"Eric  
Ries",price:750,category:"Business",quantity:8},  
  
{bookId:50,bookName:"Zero to One",author:"Peter  
Thiel",price:650,category:"Business",quantity:6}  
])
```

```

db.books.insertMany([
  {bookId:2,bookName:"Java Programming",author:"Herbert Schildt",price:700,category:"Technology",quantity:8},
  {bookId:3,bookName:"Data Science",author:"John Smith",price:800,category:"Technology",quantity:6},
  {bookId:4,bookName:"Machine Learning",author:"Tom Mitchell",price:900,category:"Technology",quantity:5},
  {bookId:5,bookName:"Atomic Habits",author:"James Clear",price:500,category:"Self Help",quantity:12},
  {bookId:6,bookName:"The Alchemist",author:"Paulo Coelho",price:300,category:"Fiction",quantity:7},
  {bookId:7,bookName:"Rich Dad Poor Dad",author:"Robert Kiyosaki",price:450,category:"Finance",quantity:9},
  {bookId:8,bookName:"Ikigai",author:"Hector Garcia",price:350,category:"Self Help",quantity:11},
  {bookId:9,bookName:"Harry Potter",author:"J.K. Rowling",price:600,category:"Fantasy",quantity:6},
  {bookId:10,bookName:"Wings of Fire",author:"A.P.J. Abdul Kalam",price:250,category:"Biography",quantity:10},
  {bookId:11,bookName:"Clean Code",author:"Robert Martin",price:850,category:"Technology",quantity:5},
  {bookId:12,bookName:"C Programming",author:"Dennis Ritchie",price:400,category:"Technology",quantity:8},
  {bookId:13,bookName:"Think and Grow Rich",author:"Napoleon Hill",price:300,category:"Self Help",quantity:9},
  {bookId:14,bookName:"The Hobbit",author:"J.R.R. Tolkien",price:550,category:"Fantasy",quantity:7},
  {bookId:15,bookName:"1984",author:"George Orwell",price:450,category:"Fiction",quantity:6},
  {bookId:16,bookName:"SQL Fundamentals",author:"John Watson",price:650,category:"Technology",quantity:8},
  {bookId:17,bookName:"Deep Work",author:"Cal Newport",price:500,category:"Self Help",quantity:10},
  {bookId:18,bookName:"Atomic Structure",author:"R. Sharma",price:200,category:"Education",quantity:12},
  {bookId:19,bookName:"Digital Marketing",author:"Philip Kotler",price:750,category:"Business",quantity:5},
  {bookId:20,bookName:"Business Studies",author:"Poonam Gandhi",price:350,category:"Business",quantity:8},
  {bookId:21,bookName:"Operating Systems",author:"Galvin",price:800,category:"Technology",quantity:6},
  {bookId:22,bookName:"Computer Networks",author:"Andrew Tanenbaum",price:850,category:"Technology",quantity:5},
  {bookId:23,bookName:"Database Systems",author:"Raghu Ramakrishnan",price:900,category:"Technology",quantity:4},
  {bookId:24,bookName:"Artificial Intelligence",author:"Stuart Russell",price:950,category:"Technology",quantity:5},
  {bookId:25,bookName:"The Psychology of Money",author:"Morgan Housel",price:600,category:"Finance",quantity:9},
  {bookId:26,bookName:"Rich Dad Poor Dad 2",author:"Robert Kiyosaki",price:450,category:"Finance",quantity:7},
  {bookId:27,bookName:"Sherlock Holmes",author:"Arthur Conan Doyle",price:300,category:"Fiction",quantity:6},
  {bookId:28,bookName:"The Kite Runner",author:"Khaled Hosseini",price:400,category:"Fiction",quantity:5}
])

```

4. DISPLAY ALL STUDENTS

MongoDB Query:

```
db.books.find()
```

Explanation: Retrieves documents from the books collection.

```
> db.books.find()
{
  _id: ObjectId('6ab295f279f427617fd6af9f'),
  bookId: 1,
  bookName: 'Python Basics',
  author: 'Mark Lutz',
  price: 500,
  category: 'Technology',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa0'),
  bookId: 2,
  bookName: 'Java Programming',
  author: 'Herbert Schildt',
  price: 700,
  category: 'Technology',
  quantity: 8
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
```

5. EQUAL TO — \$eq

MongoDB Query:

```
db.books.find({price: {$eq: 500}})
```

Explanation: Finds books whose price is equal to 500.

```
> db.books.find({price: {$eq: 500}})
< {
  _id: ObjectId('6ab295f279f427617fd6af9f'),
  bookId: 1,
  bookName: 'Python Basics',
  author: 'Mark Lutz',
  price: 500,
  category: 'Technology',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa3'),
  bookId: 5,
  bookName: 'Atomic Habits',
  author: 'James Clear',
  price: 500,
  category: 'Self Help',
  quantity: 12
}
{
  _id: ObjectId('6ab2971a79f427617fd6afaf'),
  bookId: 17,
  bookName: 'Deep Work',
  author: 'Cal Newport',
  price: 500,
  category: 'Self Help',
  quantity: 10
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.books.find({price: {$gt: 700}})
```

Explanation: Finds books whose price is greater than 700.

```
> db.books.find({price: {$gt: 700}})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa2'),
  bookId: 4,
  bookName: 'Machine Learning',
  author: 'Tom Mitchell',
  price: 900,
  category: 'Technology',
  quantity: 5
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa9'),
  bookId: 11,
  bookName: 'Clean Code',
  author: 'Robert Martin',
  price: 850,
  category: 'Technology',
  quantity: 5
}
```

7. LESS THAN — \$lt

MongoDB Query:

```
db.books.find({price: {$lt: 300}})
```

Explanation: Finds books whose price is less than 300.

```
> db.books.find({price: {$lt: 300}})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa8'),
  bookId: 10,
  bookName: 'Wings of Fire',
  author: 'A.P.J. Abdul Kalam',
  price: 250,
  category: 'Biography',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afb0'),
  bookId: 18,
  bookName: 'Atomic Structure',
  author: 'R. Sharma',
  price: 200,
  category: 'Education',
  quantity: 12
}
{
  _id: ObjectId('6ab2971a79f427617fd6afc2'),
  bookId: 36,
  bookName: 'Statistics Basics',
  author: 'S.C. Gupta',
  price: 250,
  category: 'Education',
  quantity: 10
}
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.books.find({price: {$gte: 800}})
```

Explanation: Finds books whose price is greater than or equal to 800.

```
> db.books.find({price: {$gte: 800}})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa2'),
  bookId: 4,
  bookName: 'Machine Learning',
  author: 'Tom Mitchell',
  price: 900,
  category: 'Technology',
  quantity: 5
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa9'),
  bookId: 11,
  bookName: 'Clean Code',
  author: 'Robert Martin',
  price: 850,
  category: 'Technology',
  quantity: 5
}
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.books.find({price: {$lte: 400}})
```

Explanation: Finds books whose price is less than or equal to 400.

```
> db.books.find({price: {$lte: 400}})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa4'),
  bookId: 6,
  bookName: 'The Alchemist',
  author: 'Paulo Coelho',
  price: 300,
  category: 'Fiction',
  quantity: 7
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa6'),
  bookId: 8,
  bookName: 'Ikigai',
  author: 'Hector Garcia',
  price: 350,
  category: 'Self Help',
  quantity: 11
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa8'),
  bookId: 10,
  bookName: 'Wings of Fire',
  author: 'A.P.J. Abdul Kalam',
  price: 250,
  category: 'Biography',
  quantity: 10
}
```

10. UPDATE OPERATION

MongoDB Query:

```
db.books.updateOne({bookId:5}, {$set:{price:600}})
```

Explanation: Updates the price of the book with bookId 5 to 600.

```
> db.books.updateOne({bookId:5}, {$set:{price:600}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

11. DELETE ONE OPERATION

MongoDB Query:

```
db.books.deleteOne({bookId:10})
```

Explanation: Deletes the book with bookId 10.

```
> db.books.deleteOne({bookId:10})
{
  acknowledged: true,
  deletedCount: 1
}
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.books.deleteMany({price: {$lt: 300}})
```

Explanation: Deletes all books whose price is less than 300.

```
> db.books.deleteMany({price: {$lt: 300}})
< {
  acknowledged: true,
  deletedCount: 2
}
```

13. AND OPERATION — \$and

MongoDB Query:

```
db.books.find({$and:[{price:{$gt:500}},{category:"Self Help"}]})
```

Explanation: Finds books with price greater than 500 and category Self Help.

```
> db.books.find({$and:[{price:{$gt:500}},{category:"Self Help"}]})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa3'),
  bookId: 5,
  bookName: 'Atomic Habits',
  author: 'James Clear',
  price: 600,
  category: 'Self Help',
  quantity: 12
}
{
  _id: ObjectId('6ab2971a79f427617fd6afbc'),
  bookId: 30,
  bookName: 'The Power of Now',
  author: 'Eckhart Tolle',
}
```

14. OR OPERATION — \$or

MongoDB Query:

```
db.books.find({$or:[{category:"Fiction"},{category:"Fantasy"}]})
```

Explanation: Finds books whose category is Fiction or Fantasy.

```
db.books.find({$or:[{category:"Fiction"},{category:"Fantasy"}]})
{
  _id: ObjectId('6ab2971a79f427617fd6afa4'),
  bookId: 6,
  bookName: 'The Alchemist',
  author: 'Paulo Coelho',
  price: 300,
  category: 'Fiction',
  quantity: 7
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa7'),
  bookId: 9,
  bookName: 'Harry Potter',
  author: 'J.K. Rowling',
  price: 600,
  category: 'Fantasy',
  quantity: 6
}
{
  _id: ObjectId('6ab2971a79f427617fd6afac'),
  bookId: 14,
  bookName: 'The Hobbit',
  author: 'J.R.R. Tolkien',
  price: 550,
  category: 'Fantasy',
  quantity: 7
}
```

15. SORT OPERATION

MongoDB Query:

```
db.books.find().sort({price:1})
```

Explanation: Sorts books by price in ascending order.

```
> db.books.find().sort({price:1})
< {
  _id: ObjectId('6ab2971a79f427617fd6afa4'),
  bookId: 6,
  bookName: 'The Alchemist',
  author: 'Paulo Coelho',
  price: 300,
  category: 'Fiction',
  quantity: 7
}
{
  _id: ObjectId('6ab2971a79f427617fd6afab'),
  bookId: 13,
  bookName: 'Think and Grow Rich',
  author: 'Napoleon Hill',
  price: 300,
  category: 'Self Help',
  quantity: 9
}
{
  _id: ObjectId('6ab2971a79f427617fd6afb9'),
  bookId: 27,
  bookName: 'Sherlock Holmes',
  author: 'Arthur Conan Doyle',
  price: 300
}
```

B. MongoDB Query:

```
b.books.find().sort({price:-1})
```

Explanation: Sorts books by price in descending order.

```
> db.books.find().sort({price:-1})
< {
  _id: ObjectId('6ab2971a79f427617fd6afb6'),
  bookId: 24,
  bookName: 'Artificial Intelligence',
  author: 'Stuart Russell',
  price: 950,
  category: 'Technology',
  quantity: 5
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa2'),
  bookId: 4,
  bookName: 'Machine Learning',
  author: 'Tom Mitchell',
  price: 900,
  category: 'Technology',
  quantity: 5
}
{
  _id: ObjectId('6ab2971a79f427617fd6afb5'),
  bookId: 23,
  bookName: 'Database Systems',
  author: 'Raghu Ramakrishnan',
  price: 900,
  category: 'Technology',
  quantity: 4
}
```

16. LIMIT OPERATION

MongoDB Query:

```
db.books.find().limit(5)
```

Explanation: Displays only the first 5 books.

```
>_MONGOSH
> db.books.find().limit(5)
< {
  _id: ObjectId('6ab295f279f427617fd6af9f'),
  bookId: 1,
  bookName: 'Python Basics',
  author: 'Mark Lutz',
  price: 500,
  category: 'Technology',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa0'),
  bookId: 2,
  bookName: 'Java Programming',
  author: 'Herbert Schildt',
  price: 700,
  category: 'Technology',
  quantity: 8
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
```

17. SEARCH BY CITY

MongoDB Query:

```
db.books.find({category:"Technology"})
```

Explanation: Finds all books belonging to the Technology category.

```
>_MONGOSH
>
> db.books.find({category:"Technology"})
< {
  _id: ObjectId('6ab295f279f427617fd6af9f'),
  bookId: 1,
  bookName: 'Python Basics',
  author: 'Mark Lutz',
  price: 500,
  category: 'Technology',
  quantity: 10
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa0'),
  bookId: 2,
  bookName: 'Java Programming',
  author: 'Herbert Schildt',
  price: 700,
  category: 'Technology',
  quantity: 8
}
{
  _id: ObjectId('6ab2971a79f427617fd6afa1'),
  bookId: 3,
  bookName: 'Data Science',
  author: 'John Smith',
  price: 800,
  category: 'Technology',
  quantity: 6
}
```

18. COUNT DOCUMENTS

MongoDB Query:

```
db.books.countDocuments()
```

Explanation: Counts the total number of books in the collection.

```
> db.books.countDocuments()
```

```
< 47
```

```
library> |
```

MONGODB OPERATORS USED:

Operator	Meaning	Example
\$eq	Equal to	marks: {\$eq:88}
\$gt	Greater than	marks: {\$gt:85}
\$lt	Less than	marks: {\$lt:60}
\$gte	Greater than or equal to	marks: {\$gte:90}
\$lte	Less than or equal to	marks: {\$lte:70}
\$and	All conditions true	\$and:[...]
\$or	At least one condition true	\$or:[...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to Ascending order & descending order data, comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.